

SDVoE Control Server SDVoE API

Software Release Notes Version 3.9.0.1

Revision History

Version	Release Date	API Syntax & Control Server Revisions
3.9.0.1	February 2026	SDVoE API and Control Server 3.9.0.1
3.9	December 2025	SDVoE API and Control Server 3.9.0.0
3.8	March 2025	SDVoE API and Control Server 3.8.0.1 Added support of Semtech UEX USB 2.0 extender.
3.7	April 2024	Document version number changed to match API version. SDVoE API 3.7 and SDVoE Control Server 3.7.0.0.
1.9	October 2023	For feature “Set device name to its mac address” version referred to was 1.5 but should have been 2.0.
1.8	September 2023	SDVoE API 3.6 and SDVoE Control Server 3.6.0.0
1.7	February 2023	SDVoE API 3.5 and SDVoE Control Server 3.5.0.0
1.6	May 2022	SDVoE API 3.4 and SDVoE Control Server 3.4.0.0
1.5	January 2022	SDVoE API 3.3 and SDVoE Control Server 3.3.1.0
1.4	October 2021	SDVoE API 3.3 and SDVoE Control Server 3.3.0.0
1.3	July 2021	SDVoE API 3.2 and SDVoE Control Server 3.2.0.0 release
1.2	November 2020	BlueRiver Control Server 3.1.1.0 release.
1.1	July 2020	BlueRiver API 3.1 and BlueRiver Control Server 3.1.0.0 release
1.0	May 2020	BlueRiver API 3.0 and BlueRiver Control Server 3.0.0.0 release

Table of Contents

1	INTRODUCTION.....	5
1.1	PLATFORM SUPPORT	5
1.2	CHIPSET SUPPORT	5
1.3	RELEASE OVERVIEW.....	6
2	NEW FUNCTIONALITY	7
3	FIXED ISSUES	8
4	KNOWN ISSUES	9
5	PREVIOUS FIRMWARE RELEASE NOTES	10
5.1	VERSION 3.9.0.0 RELEASE NOTES	10
5.1.1	<i>New Functionality.....</i>	10
5.1.2	<i>Fixed Issues.....</i>	11
5.2	VERSION 3.8.0.1 RELEASE NOTES	11
5.2.1	<i>New Functionality.....</i>	11
5.2.2	<i>Fixed Issues.....</i>	12
5.3	VERSION 3.7.0.0 RELEASE NOTES	12
5.3.1	<i>New Functionality.....</i>	12
5.3.2	<i>Fixed Issues.....</i>	14
5.4	VERSION 3.6.0.0 RELEASE NOTES	14
5.4.1	<i>New Functionality.....</i>	14
5.4.2	<i>Fixed Issues.....</i>	16
5.5	VERSION 3.5.0.0 RELEASE NOTES	16
5.5.1	<i>New Functionality.....</i>	16
5.5.2	<i>Fixed Issues.....</i>	17
5.6	VERSION 3.4.0.0 RELEASE NOTES	17
5.6.1	<i>New Functionality.....</i>	18
5.6.2	<i>End of API 2.xx Support</i>	18
5.6.3	<i>Fixed Issues.....</i>	18
5.7	VERSION 3.3.1.0 RELEASE NOTES	19
5.7.1	<i>New Functionality.....</i>	19
5.7.2	<i>End of API 2.xx Support Reminder.....</i>	19
5.7.3	<i>Fixed Issues.....</i>	20
5.8	VERSION 3.3.0.0 RELEASE NOTES	20
5.8.1	<i>New Functionality.....</i>	20
5.8.2	<i>End of API 2.xx support</i>	21
5.8.3	<i>Fixed Issues.....</i>	21
5.9	VERSION 3.2.0.0 RELEASE NOTES	21
5.9.1	<i>New Functionality.....</i>	22
5.9.2	<i>Fixed Issues.....</i>	22
5.10	VERSION 3.1.1.0 RELEASE NOTES	22
5.10.1	<i>New Functionality.....</i>	23
5.10.2	<i>Fixed Issues.....</i>	23

5.11	VERSION 3.1.0.0 RELEASE NOTES	23
5.11.1	<i>New Functionality</i>	23
5.12	VERSION 3.0.0.0 RELEASE NOTES	24
5.12.1	<i>New Functionality</i>	24
5.12.2	<i>Back compatibility mode</i>	25

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1 Introduction

These release notes are for the SDVoE Control Server and SDVoE API.

The Control Server implements the SDVoE API. The server forms an abstraction layer between the AV application software and SDVoE endpoints. Rather than accessing hardware directly, the control application interacts with the Control Server through SDVoE API commands. The control application is therefore hardware agnostic and interoperable with all BlueRiver® endpoints and edge devices from any hardware vendor and manufacturer.

1.1 Platform Support

SDVoE Control Server binaries are available for the following platforms:

- X86 (Intel and AMD)
 - Windows 7 or later, 64-bit
 - Linux 32-bit / Linux 64-bit
- ARMv4 or later – gcc 494
 - Linux 32-bit soft-float ABI (using gcc 4.9.4)
- ARMv4 or later
 - Linux 32-bit soft-float ABI (using gcc 8.3.0)
- ARMv6 or later with VFP (Vector Floating Point extension) – gcc 494
 - Linux 32-bit hard-float ABI (e.g. Debian/Raspbian) (using gcc 4.9.4)
- ARMv6 or later with VFP (Vector Floating Point extension)
 - Linux 32-bit hard-float ABI (e.g. Debian/Raspbian) (using gcc 8.3.0)
- ARMv8 Aarch64 (ARM64) – gcc 5.5
 - Linux 64-bit (using gcc 5.5). Available since version 3.6.
- ARMv8 Aarch64 (ARM64)
 - Linux 64-bit (using gcc 9.3). Available since API version 3.2.0.1.
- Apple Mac X86
 - MacOS 10.12 or later.
- Apple Mac M1 Aarch64 (ARM64)
 - MacOS 10.12 or later. Available since API version 3.5.
- AndroidNative binary for ARMv7 32-bit and ARMv7 64-bit. Available since API version 3.4.

1.2 Chipset Support

SDVoE Control Server and API support the following chipsets:

- AVP1000 BlueRiver AV Processor
- AVP2000 BlueRiver AV Processor
- AVP2000T BlueRiver AV Processor
- NT1000 BlueRiver AV Processor
- NT2000 BlueRiver AV Processor
- UEX1000 USB 2.0 Extender
- UEX2000 USB 2.0 Extender

1.3 Release Overview

Release Date: February 2026

This is a general hotfix release, including bug fix. This release does not support 2.xx syntax nor USB Icron syntax. All applications should be updated to API 3.x syntax as of the 3.4 release and to generic USB Extender syntax as of 3.9 release.

This release includes the following:

Name	Version	Comments
SDVoE API and Control Server	3.9.0.1	General hotfix release.

This release supports BlueRiver firmware as listed below or older:

Chipset Series	FW	Comments
BlueRiver AV Processor	2.4.0.1	AVP1000, AVP2000 and AVP2000T
BlueRiver NT2000/NT1000	4.1.2.0	NT2000 and NT1000

This release supports UEX firmware as listed below or older:

Chipset Series	FW	Comments
Semtech UEX2000/UEX1000	1.2.0.0	UEX2000 and UEX1000

This release supports BlueRiver Bitmap Streamer software as listed below or older:

Software Device	SW	Comments
Semtech BlueRiver Bitmap Streamer	1.1.0.0	BlueRiver Bitmap Streamer

IMPORTANT! When upgrading hardware firmware recommended to always upgrade the SDVoE Control Server to a matching revision to ensure new functionality is supported.

2 New Functionality

No new functionality included in this SDVoE Control Server and SDVoE API version compared to the previous 3.9.0.0 general release.

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3 Fixed Issues

This release fixes the following issues from previous API Server release.

Issue	Description
NTP-20342	Cannot set AVP static IP address when Icron is installed and API Icron control is disabled.

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4 Known Issues

This release has the following known issues.

Issue	Description
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5 Previous Firmware Release Notes

The following sections list the content of previous release notes.

5.1 Version 3.9.0.0 Release Notes

Release Date: December 2025

This is a new feature release. It includes the new features listed in section 5.1.1 New Functionality, along with a handful of internal bug fixes. This release does not support 2.xx syntax nor USB Icron syntax. All applications should be updated to API 3.x syntax as of the 3.4 release and to generic USB Extender syntax as of 3.9 release.

This release includes the following:

Name	Version	Comments
SDVoE Control Server	3.9.0.0	General release.
SDVoE API	3.9	

This release supports BlueRiver firmware as listed below or older:

Chipset Series	FW	Comments
BlueRiver AV Processor	2.4.0.0	AVP1000, AVP2000 and AVP2000T
BlueRiver NT2000/NT1000	4.1.2.0	NT2000 and NT1000

This release supports UEX firmware as listed below or older:

Chipset Series	FW	Comments
Semtech UEX2000/UEX1000	1.2.0.0	UEX2000 and UEX1000

This release supports BlueRiver Bitmap Streamer software as listed below or older:

Software Device	SW	Comments
Semtech BlueRiver Bitmap Streamer	1.1.0.0	BlueRiver Bitmap Streamer

IMPORTANT! When upgrading hardware firmware recommended to always upgrade the SDVoE Control Server to a matching revision to ensure new functionality is supported.

5.1.1 New Functionality

The following new functionality is now included in SDVoE Control Server and SDVoE API.

- Support the control via API of up to 2 UEX (UEX1000 or UEX2000) USB 2.0 extenders attached to a single AVP chip.
- Support video freeze/unfreeze in fast switch via API command, allowing to switch the HDMI input video source without introducing black frames during transition
- Added reporting of the HDMI output monitor (sink device) HDCP level support
- Added reporting of AVP core dump
- Removed USB Icron syntax support, see PDS-063049 SDVoE API Icron to USB Extender Migration Guide.

5.1.2 Fixed Issues

This release fixes the following issues from previous API Server release.

Issue	Description
NTP-20040	Cannot firmware upgrade Audio Bridge (regression in API 3.8.0.1)

5.2 Version 3.8.0.1 Release Notes

Release Date: April 2024

This release includes the following:

Name	Version	Comments
SDVoE Control Server	3.8.0.1	General release.
SDVoE API	3.8	

This release supports BlueRiver firmware as listed below or older:

Chipset Series	FW	Comments
BlueRiver AV Processor	2.3.0.1	AVP1000, AVP2000 and AVP2000T
BlueRiver NT2000/NT1000	4.1.2.0	NT2000 and NT1000

This release supports UEX firmware as listed below or older:

Chipset Series	FW	Comments
Semtech UEX2000/UEX1000	1.0.0.2	UEX2000 and UEX1000

This release supports BlueRiver Bitmap Streamer software as listed below or older:

Software Device	SW	Comments
Semtech BlueRiver Bitmap Streamer	1.1.0.0	BlueRiver Bitmap Streamer

IMPORTANT! When upgrading hardware firmware recommended to always upgrade the SDVoE Control Server to a matching revision to ensure new functionality is supported.

5.2.1 New Functionality

The following new functionality is now included in SDVoE Control Server and SDVoE API.

- **Semtech UEX2000 and UEX1000 USB 2.0 Extender support:** Support has been added in the Control server API to discover, first-time program, firmware update and control the new UEX USB 2.0 extenders. See the following documentation for more information on these products:
 - PDS-062983_UEX1000_Product_Brief
 - PDS-062990_UEX2000_Product_Brief
 - PDS-062986_UEX_USB_Extender_DataSheet
- **Memory management improvements:** Internal code structure modifications for better performance.
- **Dynamic control of the BlueRiver Bitmap Streamer (version 1.1):** A new “update_stream” API commands has been added to the API to take advantage of the new dynamic features of the BlueRiver Bitmap Streamer added its version 1.1, allowing to load or replace a file for a certain

existing or new stream.

5.2.2 Fixed Issues

This release fixes the following issues from previous API Server release.

Issue	Description
NTP-18973	BlueRiver Bitmap Streamer fix join to already joined stream creates a glitch (BBS 1.1.0.0)
NTP-18974	BlueRiver Bitmap Streamer start command can result in device timeout (BBS 1.1.0.0)
NTP-18975	BlueRiver Bitmap Streamer momentary image artifact long term steady (BBS 1.1.0.0)
NTP-19352	Added API error to report if EDID read attempt return is an invalid EDID
NTP-19429	Added API error to prevent user from sending USB-HID switch to devices with same role
NTP-19545	Cannot reset network settings on ICRON if network settings are wrong

5.3 Version 3.7.0.0 Release Notes

Release Date: September 18, 2023

This release includes the following:

Name	Version	Comments
SDVoE Control Server	3.7.0.0	General release.
SDVoE API	3.7	

This release supports BlueRiver firmware as listed below or older:

Chipset Series	FW	Comments
BlueRiver AV Processor	2.2.0.0	AVP1000, AVP2000 and AVP2000T
BlueRiver NT2000/NT1000	4.1.2.0	NT2000 and NT1000

IMPORTANT! When upgrading hardware firmware recommended to always upgrade the SDVoE Control Server to a matching revision to ensure new functionality is supported.

5.3.1 New Functionality

The following new functionality is now included in SDVoE Control Server and SDVoE API.

- **API redundancy mode:** A redundancy mode has been added to the API configuration file, to help customers implement API redundancy in their system. This redundancy mode, when enabled, puts the API in an idle mode until an external connection occurs. This allows multiple API on the network without the undesired extra traffic of all those API requesting info from the devices at the same time. Only the one that is enabled by the user opening a connection will request info from devices, the others will stay idle waiting to be awakened by external connection when the primary API is down for example. Redundancy management still needs to be handled by the customers system.
- **API memory management improvements:** A maximum threshold configuration for pending device requests has been added in the API configuration file, to allow API admin to specify the threshold of user requests at which the API would return an error rather than filling up its memory when user is requesting more than the API server memory is able to handle. A pending device request abort on connection close configuration has been added in the API configuration

file for the API to release the request when the connection that made the request is closed (therefore not needed anymore). Configuration has also been added in the API configuration file to allow control over lifetime of completed events based on expiration. Other internal memory management improvements have also been implemented.

- **API commands to get last event and/or time:** New event and timestamp commands have been added to only get the last event ID and time or only the time without requesting an event.
- **Detect/recover when change in Ethernet Adapter:** A network monitor enable configuration has been added to the API configuration file to monitor change in Ethernet adapters and be able to recover from connection lost or new connection while ensuring not to drop the current connection.
- **API interfaces IP info:** An interface section listing the IP addresses of the API server network interfaces has been added to the API JSON return.
- **BlueRiver Bitmap Streamer support:** Support has been added for BlueRiver Bitmap Streamer software devices (v1.0.0.0) to be detected as SDVoE endpoints by the API and allow starting/stopping of their streams. See the Semtech BlueRiver Bitmap Streamer User Guide (PDS-062933) for details.

The following new functionality is now included in SDVoE Control Server and SDVoE API and requires BlueRiver units to be updated with firmware version 2.2.

- **802.1X support:** 802.1X authentication support using EAP-TLS has been added in devices firmware, to allow validation of access to send/receive packets on a given network. API commands have been added to allow pushing 802.1X certificate in devices as well as retrieving info from it.
- **Send Wake-on-LAN (WoL) packet via AVP device:** An API command has been added to allow sending a WoL packet from the API to an AVP device, for the AVP device to relay it on the network (acts as a WoL proxy) by broadcast the message to the local subnet to wake a powered-off target machine.
- **HELLO packets to IP from DHCP option 43:** Up to ASIC firmware version 2.0 and API 3.5, HELLO packets were only broadcasted. From firmware version 2.1 and API version 3.6, SDVoE systems can be configured to use only multicast HELLO packets rather than broadcast. As of firmware version 2.2 and API version 3.7, support has been added to support sending the HELLO packets to unicast IP address(es) set in the DHCP server option 43. The AVP device will identify itself using DHCP option 60 when sending DHCP-DISCOVER and DHCP-REQUEST messages, with the "Semtech AVP" Vendor Class Identifier, and will expect DHCP server option 43 with sub-option 1 to convey a list of API IP address(es). Broadcast HELLO packets are still enabled by default and may be disabled by setting `disable_device_broadcast` to 'yes' in the configuration file. However, if the DHCP server is configured to support option 60 and option 43 and the latter one is programmed with IP address(es), this will take precedence over the multicast or broadcast configuration, therefore the FW will send unicast HELLO packets. This is only applicable of course if the IP mode of the FW is DHCP.
- **USB-HID statistics:** A remote extender statistics section has been added in the USB-HID node status of the get device/settings API command JSON return. This new section gives info on the total number of devices (including USB hubs) connected to the remote USB connector, the number of USB hubs connected, the number of supported USB devices connected, the number

of unsupported USB devices connected and the number of active endpoints per type (interrupt and bulk).

5.3.2 Fixed Issues

This release fixes the following issues from previous API Server release.

Issue	Description
NTP-17945	API not preventing user to overload system memory by making too many requests.
NTP-18091	Icron not found corner case issue.
NTP-18159	Sometimes devices show as unreachable after power off/on 10G switch or disconnect/reconnect switch to API server.
NTP-18514	OpenSSL update to latest version (3.2.1).
NTP-18521	API will not always start if started during NIC startup instability.
NTP-18603	API reports wrong authenticated status for get device but ok for get settings.
NTP-18609	Dual link join command error when only one link connected.

5.4 Version 3.6.0.0 Release Notes

Release Date: September 18, 2023

This release includes the following:

Name	Version	Comments
SDVoE Control Server	3.6.0.0	General release.
SDVoE API	3.6	

This release supports BlueRiver firmware as listed below or older:

Chipset Series	FW	Comments
BlueRiver AV Processor	2.1.0.0	AVP1000, AVP2000 and AVP2000T
BlueRiver NT2000/NT1000	4.1.2.0	NT2000 and NT1000

IMPORTANT! When upgrading hardware firmware recommended to always upgrade the SDVoE Control Server to a matching revision to ensure new functionality is supported.

5.4.1 New Functionality

The following new functionality is now included in SDVoE Control Server and SDVoE API.

- **Additional API platform:** Linux ARMv8 AArch64 gcc 5.5 has been added to the list of supported platforms. The binary is in the "linux-armv8-aarch64-gcc5" release folder. See the SDVoE Developers API Reference Guide (PDS-062489) for details.
- **RS232 HEX mode:** New HEX data format option for RS232 communication between control client and RS232 peripherals connected to endpoints. The RS232 mode, HEX or ASCII, is set with a new parameter as part of the send RS232 command, for sending RS232 data via the API. As for receiving RS232 data from the API, both ASCII and HEX data are now part of the JSON return. The default is ASCII. See the SDVoE Developers API Reference Guide (PDS-062489) for details.
- **Detect API on the network:** API instances on the network are now easily discoverable by client sessions running on the network. Specified in the configuration file, this feature is disabled by

default. When enabled, the API sends multicast discovery packets. See the SDVoE Control Server Administrator Guide (PDS-062492) and the SDVoE API Application Note Library (PDS-062491) for details.

- **Set device name to its mac address:** The set property configuration.device_name command now includes a new reserved keyword. The keyword, mac_address, sets the target's device_name to its MAC address. When using the keyword, the command may be targeted at a single device or a group of devices. Recall that as of FW version 2.0, device_name is persistent following factory reset. To reset all units' device_name to their mac address, this new feature may be used with target set to ALL. See the SDVoE Developers API Reference Guide (PDS-062489) for details.
- **Device ID and name included in subscription information:** Subscription information now include source device_id, device_name and stream-index. A new get paths command returns all source/subscription pairs for a specified device or group of devices. See the SDVoE Developers API Reference Guide (PDS-062489) for details.
- **Configurable Icron polling:** When Icron control is enabled via the SDVoE API, the SDVoE API polls the Icron devices every 15ms. This polling interval is now programable via the configuration file parameter icron_polling_rate_sec. By setting the parameter to zero, Icron polling is disabled. See the SDVoE Control Server Administrator Guide (PDS-062492) for details.

The following new functionality is now included in SDVoE Control Server and SDVoE API and requires BlueRiver units to be updated with FW version 2.1.

- **Multicast HELLO packets:** previous firmware and API release only use broadcast HELLO packets which require an IP helper in large multi-switch networks. SDVoE systems can now be configured to use only multicast HELLO packets. Broadcast HELLO packets are enabled by default and may be disabled by setting disable_device_broadcast to 'yes' in the configuration file. To maintain backward compatibility, devices running FW older than version 2.1 will continue to send broadcast HELLO packets regardless of this disable_device_broadcast configuration file setting. In the case of multiple SDVoE APIs running on a network, it is critical to update all APIs to version 3.6 or later.

WARNING: Failure to update all APIs to version 3.6 or later may result in devices disappearing and being reported as unreachable.

If the configuration file has `disable_device_broadcast = yes`, all devices on the network sending multicast HELLO packets, i.e. all devices running FW 2.1 or later, will cease sending broadcast HELLO packets. As a result, any API older than version 3.6 will no longer see these devices. See the SDVoE Control Server Administrator Guide (PDS-062492) for details.

- **Programmable CEC device addresses:** previous firmware releases have the Consumer Electronics Control logical address hard coded to 14. The release allows the CEC logical address to be configurable using the API for both the HDMI input and HDMI output CEC nodes. Refer to the SDVoE Developers API Reference Guide (PDS-062489) for details.
- **Read contents of SFP+ I²C registers:** The content of the first 128 bytes at both the SFP+ module I²C register addresses A0h and A2h are now available via dedicated I²C registers of the ASIC, and via an API request. The SFP+ module I²C registers are defined in SNIA SFF-8472-R12.4 and can be used for diagnostic purposes. Refer to the SDVoE Developers API Reference Guide

(PDS-062489) for details.

- **Control volatility of device settings:** Most device settings are non-volatile. The new configuration.volatility_settings key has been added to the API set_property command to control the volatility of these settings. Refer to the SDVoE Developers API Reference Guide (PDS-062489) for details.

5.4.2 Fixed Issues

This release fixes the following issues from previous API Server release.

Issue	Description
NTP-1929	Feature request: Detect API server(s) on the network.
NTP-16195	The Icron Node chip_preset status is reported as true even if no Icron is installed.
NTP-16519	Feature Request: RS-232 Binary Data Transfer.
NTP-16575	HDMI audio status is not cleared when there is no audio source.
NTP-16688	Feature Request: API access of SFP+ I2C registers.
NTP-17695	Multiple API on same network resulting in latency in responses.
NTP-17444	Feature Request: Make Icron polling time programable and add the option of turning it off completely.
NTP-17885	Feature Request: A single command to set the device_name of all devices to their mac address.

5.5 Version 3.5.0.0 Release Notes

Release Date: February 8, 2023

This release includes the following:

Name	Version	Comments
SDVoE Control Server	3.5.0.0	General release.
SDVoE API	3.5	No changes to the API command set.

This release supports BlueRiver firmware as listed below or older:

Chipset Series	FW	Comments
BlueRiver AV Processor	2.0.0.0	AVP1000, AVP2000 and AVP2000T
BlueRiver NT2000/NT1000	4.1.2.0	NT2000 and NT1000

IMPORTANT! When upgrading hardware firmware recommended to always upgrade the SDVoE Control Server to a matching revision to ensure new functionality is supported.

5.5.1 New Functionality

The following new functionality is now included in SDVoE Control Server and SDVoE API.

- **Native Apple M1 support:** SDVoE Control Server binaries are now available for MacOS running on the Apple M1.

The following new functionality is now included in SDVoE Control Server and SDVoE API and requires BlueRiver units to be updated with FW version 2.0.

- **Use 'Device Name' to access and control BlueRiver endpoints:** The "Device Name" property may now be used inter-changeably with "Device ID" (mac address) to access and control endpoint units.
- **API control of HDMI output +5V signal:** the state of the HDMI_TX_5V (ball C21) output can be configured using a new API command. Software can be used to override the default "always ON" setting of this output when it is desirable to completely disable the HDMI output.
- **Enable Quad I²S local loopback in TX and RX modes:** with earlier firmware releases, the QI2S local loopback can only be enabled when the device is an AVP2000T in Transceiver mode, allowing the QI2S input to be looped back to the QI2S output of the same device. This feature enables the same functionality in both TX and RX device modes.
- **Wall name for Source Independent Wall mode:** this feature allows control software to determine which endpoint devices are part of which wall configuration by simply querying the device settings. Application software can correctly detect and manage Source Independent Video walls that were created by other control applications based on the wall name.

5.5.2 Fixed Issues

This release fixes the following issues from previous API Server release.

Issue	Description
NTP-15270	Using API command to program a GPIO with an invalid index returns with SUCCESS instead of flagging an error.
NTP-15847	When programming an RX output mode using FPS to specify frame rate, the API no longer flags an error when FPS is set to zero.
NTP-16322	Get multiview command does not report the proper layout name when default compatibility_4k_2x2 name used (requires API 3.5).
NTP-16576	Sending a CEC command to ALL_RX is also reaching the HDMI input of TR units.
NTP-17002	Cannot override default Quantization Level of HDMI input using the API (requires FW 2.0).

5.6 Version 3.4.0.0 Release Notes

Release Date: May 30, 2022

This release includes the following:

Name	Version	Comments
SDVoE Control Server	3.4.0.0	Maintenance release.
SDVoE API	3.4	No changes to the API command set.

This release supports BlueRiver firmware as listed below or older:

Chipset Series	FW	Comments
BlueRiver AV Processor	1.4.0.0	AVP1000, AVP2000 and AVP2000T
Semtech AV Extender	1.4.0.0	AVX100 and AVX200T
BlueRiver NT2000/NT1000	4.1.2.0	NT2000 and NT1000

5.6.1 New Functionality

The following new functionality is now included in SDVoE Control Server and SDVoE API.

- **Android Support:** Android OS added to the list of supported platforms. A software developer can now use the added compiled binary to create an Android Image. The provided binary is compiled for **ARMV7** 32 bits CPU. It can also be used unchanged with **ARMV8 64** bits CPU. The file is statically linked with **Bionic Standard C** library for **Android 11** (API Level 30) and is back compatible with earlier versions. When creating an Android image, developers can use the binary either as a daemon or as a standalone process started by another application. For more information see the 'SDVoE Control Server Administrators Guide' (PDS-062492)
- **Icron Simultaneous User Interaction (SUI):** This release extends embedded Icron ExtremeUSB® control to include Icron Simultaneous User Interaction (SUI). SUI enables up to **seven Icron REX** devices to connect to the same Icron LEX device, which is beneficial for collaboration use cases such as training, planning conferencing, etc. The SUI feature needs to be enabled in the Icron FW and it requires **Icron FW N12 or later**. It is important to note that if the SUI feature is enabled in Icron FW, other Icron functionality such as Mass Storage Acceleration functionality has to be disabled. USB devices also need to be connected to a USB 2.0 hub. For more details, consult Icron documentation or contact Icron Support.
- **Turn Off Icron USB control:** To prevent any interference with control applications that manage and control Icron USB 2.0 devices on their own, a control parameter '**icron_ctrl_enable**' is now included inside the API Server configuration file to turn Icron control functionality **ON** or **OFF**. By default, it is **ON**.
- **AUTO_IP:** In addition to '**DHCP**' and '**Manual IP**' modes, a new mode named '**Auto IP**' mode is now supported. In this mode, BlueRiver devices always use 'Auto IP' addresses. As before, the '**set ip**' command is used to configure the IP mode. The new mode requires BlueRiver AVP chipset running FW 1.4.0.0 or later. **Important:** This new addressing mode is not supported by Icron ExtremeUSB and should not be used in a system that includes BlueRiver endpoints embedding Icron ExtremeUSB functionality.
- **Retrieve Multiview layout:** A new command '**get multiview**' is added to retrieve the current Multiview layout. Rather than keeping a log of the currently running Multiview layout settings, control applications can now query the API Server for this information. The new 'get multiview' command requires BlueRiver AVP devices running FW 1.4.0.0 or later.

5.6.2 End of API 2.xx Support

As highlighted in previous release notes, adding functionality to older generation API syntax has now ended. This API 3.4 release does not include a new 2.34 back-compatibility mode. The last compatibility version is 2.33.

To take use of new features, applications will have to support current generation (3.x) syntax. If you have not yet migrated your application to the current API syntax, now is the time to do it. For more information, see document 'BlueRiver API 3.0 Migration Guidelines (PDS-062349)'.

5.6.3 Fixed Issues

This release fixes the following issues from previous API Server release.

Issue	Description
NTP-13300	When device is switched from Wall mode to either FastSwitch or Genlock mode, KEEP values in FRAME_BUFFER node are non-zero.
NTP-13428	SETTINGS_CHANGED event not generated when changing RX HDCP mode.
NTP-14395	Issuing a command to leave multichannel subscription for ALL devices does not work. Current workaround is to issue a leave all multichannel subscriptions for all RX devices and then issue individual leave commands for every TX device
NTP-14573	When HID device is paired and its mode is changed from LOCAL or REMOTE to OTG, the device it is paired to generates a SETTINGS_CHANGED notification event every second. Current workaround is to unpair devices before changing the HID mode to OTG.
NTP-15468	Using 'set property' request to set "configuration.resume_streaming" to either TRUE or FALSE does not trigger SETTINGS_CHANGED event.
NTP-15469	When issuing 'request' command using the request_id received with DEVICE_DISCONNECTED event, the status.active is reported as TRUE instead of FALSE.

5.7 Version 3.3.1.0 Release Notes

Release Date: January 5, 2022

This release includes the following:

Name	Version	Comments
SDVoE Control Server	3.3.1.0	Maintenance release.
SDVoE API	3.3	No changes to the API command set.

This release supports BlueRiver firmware as listed below or older:

Chipset Series	FW	Comments
BlueRiver AV Processor	1.3.0.0	AVP1000, AVP2000 and AVP2000T
Semtech AV Extender	1.3.0.0	AVX100 and AVX200T
BlueRiver NT2000/NT1000	4.1.2.0	NT2000 and NT1000

IMPORTANT NOTE: When upgrading hardware firmware always make sure to also upgrade the SDVoE Control Server to corresponding revision.

5.7.1 New Functionality

This maintenance release does not include new functionality; it is a maintenance release that provides solutions for two known issues. Refer to section 5.4.2 Fixed Issues for details.

5.7.2 End of API 2.xx Support Reminder

*NOTE: This is **ONLY** applicable to control software originally written against API 2.x syntax.*

*It is **NOT** applicable to control software written against API 3.0 or newer, whether REST or TCI syntax.*

API 3.0 release introduced a handful of changes affecting control applications that were written against the previous API 2.x syntax. These changes are documented in detail in both API 3.0 release notes as well as in document BlueRiver API 3.0 Migration Guidelines (PDS-062349). To soften the burden, a **temporary** measure was put in place and since then, each 3.x API release has contained a special 2.3x compatibility mode for new functionality. For example, release 3.0 contained 2.30 compatibility mode and 3.1 contained 2.31 compatibility mode.

API 3.3 is the last API version that will contain backwards compatibility mode (2.33) for previous API generations. Although future releases will continue to support both new REST and old TCI interfaces, they will no longer support the previous generation JSON data structure.

We urge everyone that has not yet migrated their applications to API 3.x TCI or REST syntax to do so now.

5.7.3 Fixed Issues

This release fixes the following issues from previous API Server release.

Issue	Description
NTP-14962	Long term instability issues, including segmentation faults found in API Server 3.3 related to new security features.
NTP-15188	The API command "auth" (used to enable secure communication between API Server and devices) can return a false SUCCESS status when issued immediately after the "factory" command, which resets devices settings to factory defaults.
NTP-15274	Icron USB devices not always discovered.

5.8 Version 3.3.0.0 Release Notes

Release Date: October 4, 2021

This release includes the following:

Name	Version	Comments
SDVoE Control Server	3.3.0.0	Known before as BlueRiver Control Server
SDVoE API	3.3	Known before as BlueRiver API

This release supports latest BlueRiver firmware as follows:

Chipset Series	FW	Comments
BlueRiver AV Processor	1.3.0.0	AVP1000, AVP2000 and AVP2000T
Semtech AV Extender	1.3.0.0	AVX100 and AVX200T
BlueRiver NT2000/NT1000	4.1.2.0	NT2000 and NT1000

IMPORTANT NOTE: When upgrading hardware firmware always make sure to also upgrade the SDVoE Control Server to corresponding revision.

5.8.1 New Functionality

The following new functionality is now included in SDVoE Control Server and SDVoE API.

- **Icron USB 2.0 control:** Until now, all video, audio and control data routing is controlled through the SDVoE API, except for Icron USB 2.0 data which was managed directly by the control

application though the Icron UDP Control Protocol. As of this release 3.3.0.0, Icron USB control is now embedded within the SDVoE Control Server and SDVoE API as follows.

- **Switch** command is extended to establish routing between Icron USB LEX and REX devices
- **Stop** command is extended to stop and unpair Icron USB devices
- **Set IP** command has new keyword to program Icron IP address
- **Reset** command has new keyword to reset Icron device
- **ICRON_USB** node added to device model. Any changes to the node triggers a **SETTINGS_CHANGED** event

Refer to the SDVoE Developers API Reference Guide (PDS-062491) for more information.

- **Xyte HWaaS POC (proof-of-concept):** This release includes a proof of concept of capturing and uploading SDVoE device status data to the Xyte cloud server that then can be retrieved and analyzed using Xyte subscription cloud services. By default, this data gathering is turned OFF. Anyone who would like to learn more about Xyte and activate this on their system should contact Semtech support (aptovision_support@semtech.com).

5.8.2 End of API 2.xx support

API 3.0 release introduced a handful of changes affecting control applications that were written against the previous API 2.x syntax. These changes are documented in detail in both API 3.0 release notes as well as in document BlueRiver API 3.0 Migration Guidelines (PDS-062349). To soften the burden, a **temporary** measure was put in place and since then, each 3.x API release has contained a special 2.3x compatibility mode for new functionality. For example, release 3.0 contained 2.30 compatibility mode and 3.1 contained 2.31 compatibility mode.

This API 3.3 release is the last release that will contain backwards compatibility mode (2.33) for previous API generations. Although future releases will continue to support both new REST and old TCI interfaces, they will no longer support the previous generation JSON data structure.

We urge everyone that has not yet migrated their applications to API 3.x TCI or REST syntax to do so now.

5.8.3 Fixed Issues

No known previously reported issues were fixed in this release.

5.9 Version 3.2.0.0 Release Notes

Release Date: July 19, 2021

This release includes the following:

Name	Version	Comments
SDVoE Control Server	3.2.0.0	Known before as BlueRiver Control Server
SDVoE API	3.2	Known before as BlueRiver API

This release supports latest BlueRiver firmware as follows:

Chipset Series	FW	Comments
BlueRiver AV Processor	1.3.0.0	AVP1000, AVP2000 and AVP2000T
Semtech AV Extender	1.3.0.0	AVX100 and AVX200T
BlueRiver NT2000/NT1000	4.1.2.0	NT2000 and NT1000

IMPORTANT NOTE: SDVoE Control Server revision 3.2.0.0 or later is required when upgrading existing hardware to ASIC firmware 1.3.0.0.

5.9.1 New Functionality

The following new functionality is now included in SDVoE Control Server and SDVoE API.

- **Name change:** The API and Control Server is being rebranded and named after the SDVoE Alliance. This implies several changes. First of all the term '**BlueRiver**' is dropped from all file names including the control server executable, configuration file and logfile. The new names are '**controlserver.exe**', '**controlserver.conf**' and '**controlserver.log**'. The term '**BlueRiver**' is also no longer used within the updated configuration file. Finally, the use of keyword '**blueriver_api**' with '**require**' command is deprecated. The new preferred keyword is '**api**'. To facilitate existing installations, the new Control Server will still recognize and use older configuration files carrying the old '**blueriver.conf**' name. See PDS-062492 SDVoE Control Server Administrators Guide for more information.
- **Source independent wall:** This release adds new parameters to the existing '**set video**' command to describe regular matrix shape (rectangular) wall in terms of wall rows, wall columns, display index, screen resolution and bezels. Wall is now configured once and it is the device firmware instead of the application software that calculates both '**keep**' and '**viewport**' regions every time the source resolution changes. **IMPORTANT:** New wall parameters are only applicable to RX or TR endpoints using BlueRiver AVP2000 and AVP2000T chipsets. See PDS-062491 SDVoE Developers API Reference Guide and PDS-062491 SDVoE Developers API Application Note Library for more information.
- **Enhanced Network Security:** Communication between API server and endpoints can now be further secured by enabling security authentication using a 'user-defined' key. Two new commands, '**auth**' and '**unauth**' are used to authenticate or un-authenticate a device with an API Server. Implementation uses the following crypto algorithms and authentication protocols: SHA256, AES-CTR, Poly1305 and ECDH. See PDS-062491 SDVoE Developers API Reference Guide and PDS-062491 SDVoE Developers API Application Note Library for more information.
- **Complete set of audio source options for BlueRiver Transceivers:** Previous API releases omitted some audio source options related to auxiliary stereo and multichannel audio streams for BlueRiver AVP chipsets, additional options are now available. See PDS-062491 SDVoE Developers API Reference Guide and PDS-062491 SDVoE Developers API Application Note Library for more information.
- **ARM64:** In addition to ARM 32 bits support (armv4 without FP and armv6 with FP), ARM 64 bits (AArch64) is now also supported.

5.9.2 Fixed Issues

Issue	Description
NTP-12599	API can't write large USB descriptors exceeding 1024 bytes.

5.10 Version 3.1.1.0 Release Notes

Release Date: November 5, 2020

This release includes the following:

Name	Version	Comments
BlueRiver Control Server	3.1.1.0	Also known as the SDVoE Control Server
BlueRiver API	3.1	Also known as the SDVoE API

This release supports latest BlueRiver firmware as follows:

Chipset Series	FW	Comments
BlueRiver AV Processing	1.2.0.0	AVP1000, AVP2000 and AVP2000T
Semtech AV Extender	1.2.0.0	AVX100 and AVX200T
BlueRiver NT2000/NT1000	4.1.2.0	NT2000 and NT1000

IMPORTANT NOTE: BlueRiver Control Server revision 3.1.1.0 or later is required when upgrading existing hardware to ASIC firmware 1.2.0.0.

5.10.1 New Functionality

NONE. Maintenance release.

5.10.2 Fixed Issues

Issue	Description
NTP-11741	The Control Server blocks commands requesting output resolutions with a greater width than 4096 pixels. For example, requesting 5760x1080 Fast Switch resolution returns an error.
NTP-12279	'had_hdr' data member in 'VideoFormatDetails' JSON section is not always reported correctly.

5.11 Version 3.1.0.0 Release Notes

Release Date: July 31, 2020

This release includes the following:

Name	Version	Comments
BlueRiver Control Server	3.1.0.0	Also known as the API Server
BlueRiver API	3.1	Also known as the SDVoE API

This release supports latest BlueRiver firmware as follows:

BlueRiver Chipset	FW	Comments
AVP/AVX	1.1.1.0	
NT2000/NT1000	4.1.2.0	

5.11.1 New Functionality

The following new commands are now included in BlueRiver Control Server and BlueRiver API.

- **Faster Multiview bring-up [AVP]:** In order to obtain a fully subscribed and functional Multiview layout, the user has had to use command '**set multiview**' to apply the layout and then request new subscriptions for each Multiview window (surface). Command '**set multiview**' now includes new subscription keyword to specify all needed subscriptions with a single API call. By using a single command, switching to different Multiview layout now under 500msec. **NOTE:** New keyword applies only to AVP2000 and AVP2000T chipsets. Not supported for NT2000 chipset.
- **Dynamic switching of RX/TX [AVX/AVP]:** It is now possible to toggle AVP2000 and AVX100

devices between TX and RX mode. Furthermore, AVP2000T and AVX200T devices can be switched between RX, TX and TR (transceiver) modes. Both AVX and AVP devices include a new configuration node member named '**device_mode**'. Using the '**set property**' command, this member can be set to one of three possible values: TRANSMITTER, RECEIVER and TRANSCEIVER. When changing the device mode, the device will automatically reboot to apply new device configuration and settings. NOTE: New property applies only to AVX and AVP chipsets. Not supported for NT1000 and NT2000 devices.

5.12 Version 3.0.0.0 Release Notes

Release Date: March 14, 2020

This release supports all new functionality and features added to latest device firmware such as NT2000/NT1000 4.1.0.0 firmware and AVP/AVX 1.1 firmware.

This release also includes major changes for both the BlueRiver Control Server and BlueRiver API.

Name	Version	Comments
BlueRiver Control Server	3.0.0.0	Also known as the API Server
BlueRiver API	3.0	Also known as the SDVoE API

Up to now, the BlueRiver API syntax has been TXT based and API commands sent from control application to the BlueRiver Control Server over a TCP/TELNET connection. As of this release, the BlueRiver Control Server now supports a new RESTful API syntax over standard HTTP connection. Furthermore, this connection can be secured (HTTPS) through TLS certificates. This new REST/HTTP API interface greatly simplifies development of web-based control applications including mobile apps.

This release also includes modest changes to API command and return syntaxes, making both more accurate and the BlueRiver API syntax more self-explanatory and accessible to programmers that do not have a strong hardware background. Changes include name changes of some nodes and description strings in JSON returns, modification to default behavior, as well as a handful of API command syntax changes.

Due to these changes, the BlueRiver API 3.0 version breaks back-compatibility with existing control software written against API 2.x versions. For this reason, the new BlueRiver Control Server still supports older 2.x syntax. In addition, new back compatibility mode is included for existing control software to access and take use of new functionality without migrating to new API 3.0 syntax.

For more information about migrating to the new API 3.0 as well as new back compatibility mode, refer to sections below.

Note: It is strongly recommended to take use of the new RESTful API for any new application development.

5.12.1 New Functionality

The following new functionality is now included in BlueRiver Control Server:

- **RESTful API.** New API interface in addition to current Text Command Interface (TCI). **Representational state transfer (REST)** is a modern software architectural style that uses standard Web services such as HTTP and JSON to implement application interfaces and computer interactions over network. The request-response and stateless nature of a RESTful API makes it ill suited for asynchronous (i.e. server push) notifications. For this reason, the BlueRiver Control Server supports WebSocket connections. The new REST API as well as current

Text Command Interface (TCI) are enabled by default. This behaviour is configurable through the BlueRiver Control Server configuration file.

- **TLS security.** Optionally, the REST API interface can be secured using TLS. **Transport Layer Security (TLS)** is a protocol layer for securing electronic communications and widely used today with HTTP on the internet to secure access to websites (HTTPS). TLS connections are cryptographically encrypted and use cryptographic certificates for authentication. Any TLS software including the BlueRiver Control Server has a list of certificates that it trusts. These are known as **root certificate authority (CA)** certificates. In addition to server certificates, BlueRiver Control Server also supports client certificates. Semtech does not supply CA Certificates. Instead providing and configuring CA certificates are the joint responsibility of the end user and installer. Refer to BlueRiver Control Administrators Guide (PDS-062327) for more information on the setup and configuration of TLS certificates.
- **Explicit reporting of device capabilities.** New reporting has been added to make it simpler for control software to determine supported endpoint functionality and capabilities. Previously, control software needed to parse the '**get devices**' return and determine device capabilities based on absence/presence of various function sections and hardware nodes. Device capabilities can now be retrieved by any of the following methods:
 - a. Using the new '**get capabilities**' command.
 - b. The new capabilities section in '**get device**' return.
 - c. The return associated with **DEVICE_INFO_AVAILABLE** event.
- **Declaration of UART usage.** BlueRiver devices can support up to four UART interfaces. They are used to either implement an external RS 232 port or to connect and communicate with another internal subsystem. Up to now this has not been reported through the API thus developers and users needed to know from the hardware vendor which UART interface they should use to access the external RS 232 port. UART usage is now reported through the API. For each available UART, it is either reported as 'Exposed' when it is connected to external RS 232 port or 'Reserved' when used internally.
- **P2P pairing identifier.** For Semtech AVX extension devices as well as for BlueRiver AVP devices operating in Point-to-point mode, the API now reports the 'device id' of the remote device. Does not apply to BlueRiver NT2000 and NT1000.

5.12.2 Back compatibility mode

Due to the compatibility-breaking changes in API 3.0 syntax, the BlueRiver Control Server still supports previous API 2.x versions. Furthermore, it also includes a special compatibility mode for customers who have not yet been able to adjust to the updated API syntax but at the same time do need or want to take use of some of the new features.

To initiate a session, control software uses the API 'require' command. With this command, a client declares which API version it is designed for. For example if client requires API version 2.21, then it can use functionality and syntax that is included in version 2.21 or earlier. Similarly, software clients that are ready to use new API syntax and new functionality, they use the 'require' function specifying the API version 3.0.

For clients who want to use new functionality that is part of API 3.0 without being exposed to associated syntax changes, they can initiate the session by declaring version **2.30** instead of 3.0.

Semtech will maintain and keep this mode up to date until end of calendar year 2020. In that period for

every API 3.x release, it will include API 2.3x compatibility mode to access the new functionality.

When using this back-compatibility mode, it is important to keep the following in mind:

- Only applicable to existing client software initially coded against API 2.x syntax.
- Access to new RESTful API interface is not available. Anyone starting with REST will code against API 3.0 syntax.
- Access to secured TLS session is not available. TLS is only applicable to REST API.
- Semtech will stop adding new features to 2.3x compatibility mode in first quarter of 2021. Therefore, everyone should plan to adapt to the new 3.x syntax sooner than later. Changes are limited and the estimated effort to convert existing software to the new API 3.x syntax is no longer than ~1 man week.

Semtech Confidential Information
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